

Retrieval Augmented Generation (RAG) & Agent System

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Contents

- **Build a simple AI chatbot**
- **Retrieval Augmented Generation (RAG)**
- **LLM Agents**

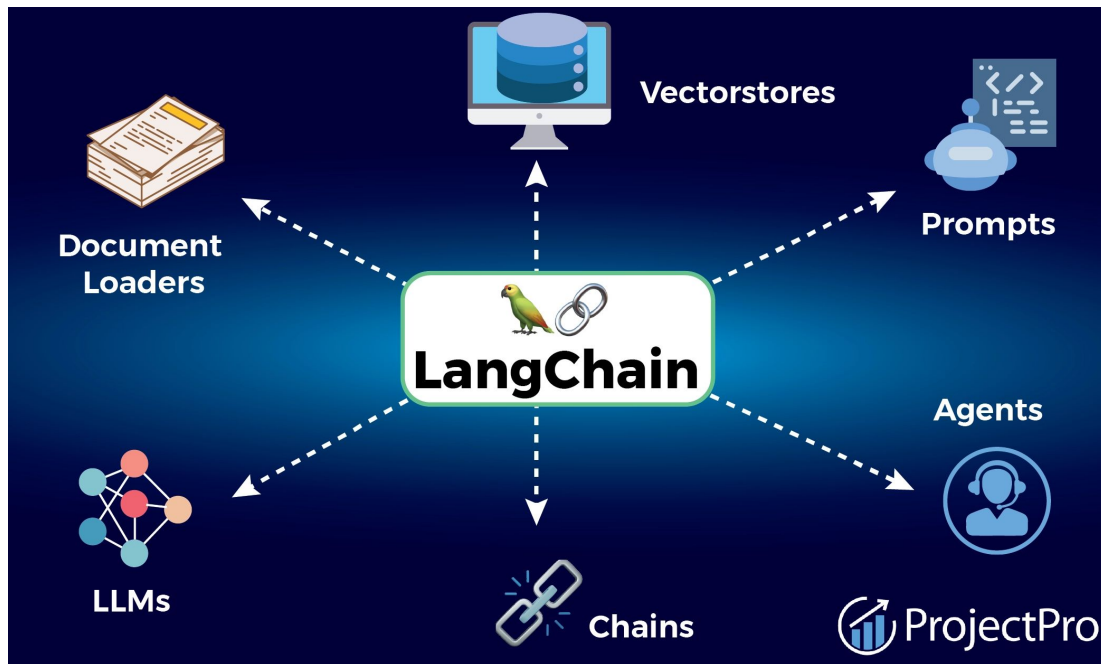
Goal of this lecture

- **Learn how to use langchain and to build an AI chatbot**
- **Understand the basics of retrieval augmented generation (RAG) and learn how to build a RAG system**
- **Understand the basic concepts of LLM agents**

Build a simple AI chatbot

What is Langchain?

- **LangChain** is a framework for developing applications powered by frontier AI models
 - <https://python.langchain.com/docs/introduction/>



Model inference

- **API inference**

- Groq (<https://groq.com/>) , Openrouter (<https://openrouter.ai/>), etc

- **Local server inference**

- vLLM (<https://docs.vllm.ai/en/latest/>)

The Unified Interface For LLMs

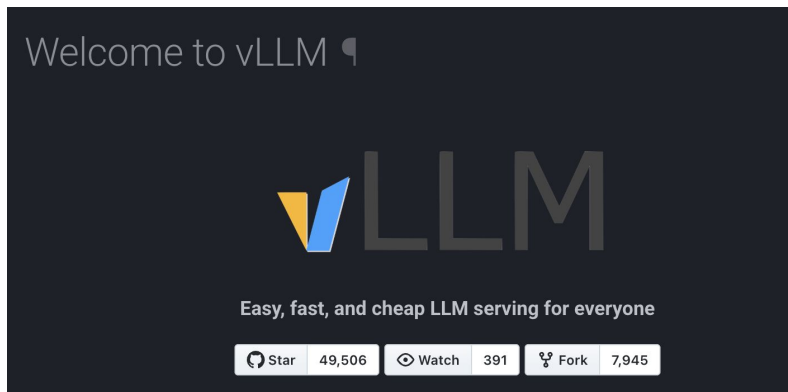
Better **prices**, better **uptime**, no subscription.

Featured Models

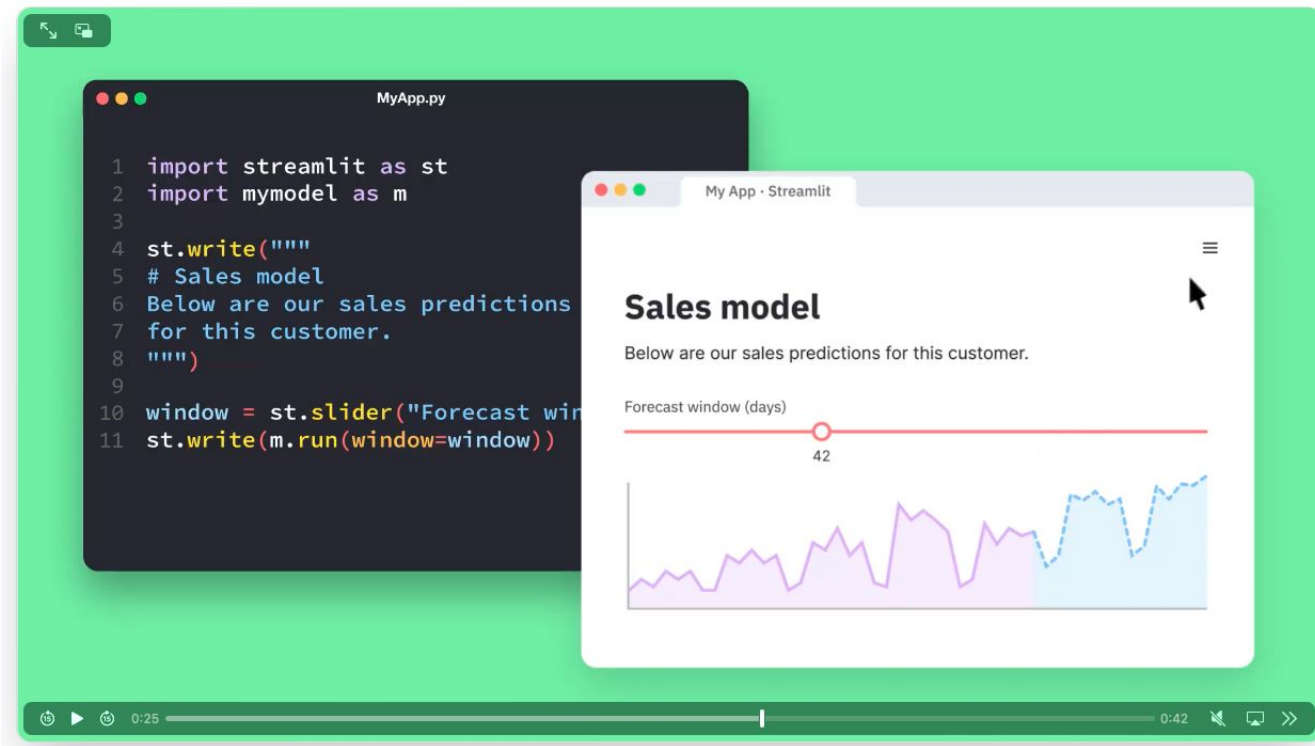
[View Trending](#)

Gemini 2.5 Pro Preview 06-05 by google		
15.4B Tokens/wk	2.2s Latency	-- Weekly growth
GPT-4.1 by openai		
47.0B Tokens/wk	771ms Latency	-1.71% Weekly growth
Claude Sonnet 4 by anthropic		
214.4B Tokens/wk	1.6s Latency	-18.11% Weekly growth



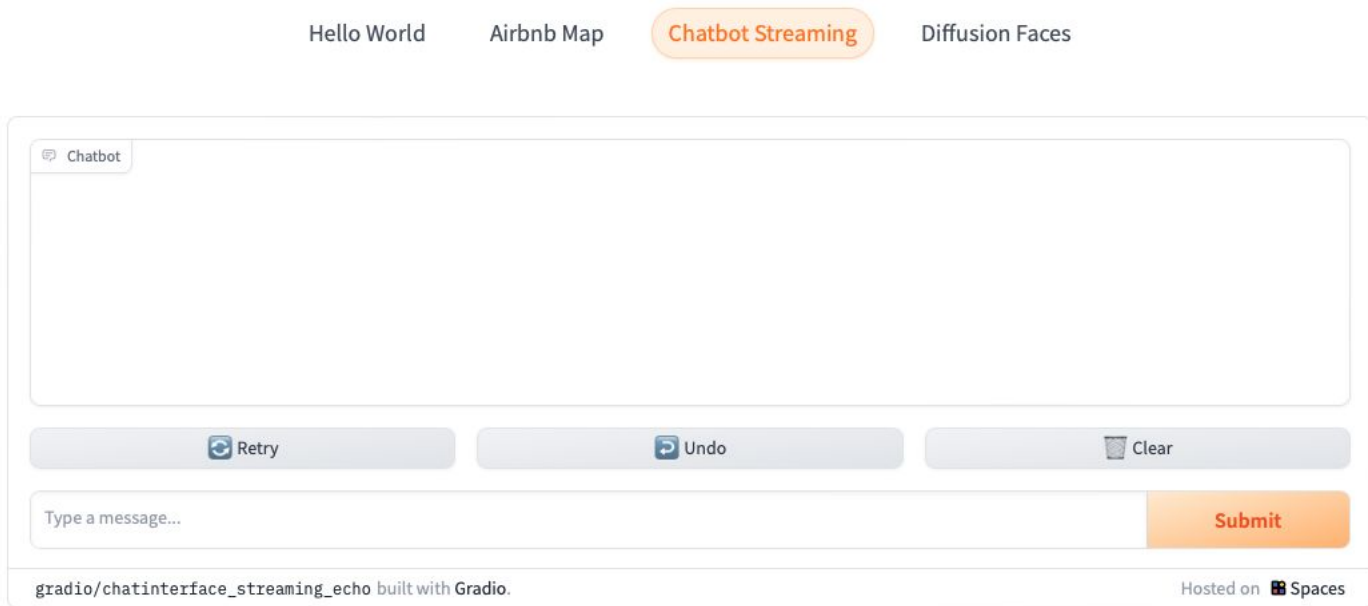
Streamlit

- <https://streamlit.io/>



Gradio

- <https://www.gradio.app/>



Tips for prompt engineering

- **Terminologies**

- **Instruction** - a specific task or instruction you want the model to perform
- **Context** - external information or additional context that can steer the model to better responses
- **Input data** - the input or question that we are interested to find a response for
- **Output indicator** - the type or format of the output

- Use some clear separator like “###” or “[]”

```
### Instruction ###  
Translate the text below to Spanish:  
Text: "hello!"
```

- It's often better to be specific and direct
- Avoid saying what **not** to do but say what to do instead

Hands On

- Chatbot with Visualization

- AI Tutor Chatbot

- 영어 문법 교정을 잡아주는 AI Tutor
 - **Scenario** : 유저는 틀린 영어 문장을 input으로 넣음
 - **Requirements** : AI Tutor는 (1) 수정된 문과 (2) paraphrase 한 문장 두개를 유저에게 제공해야함

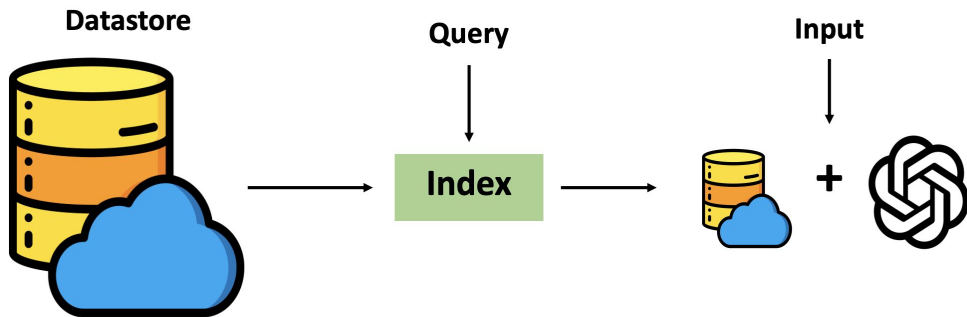
- Twenty Questions Game AI Chatbot

- 스무고개 게임을 하는 AI Chatbot
 - **Scenario** : AI가 예상하는 동물을 유저가 맞추는 게임을 진행하는 Chatbot, 유저의 질문에 따라 예, 아니요로 답해야 함
 - **Requirements** : 유저가 답이 뭔지 물어도 (20번이 되도) 절대 답을 알려주면 안됨

Retrieval Augmented Generation (RAG)

Retrieval-based language models

- Information Retrieval + Language Models
 - Retrieve information from external datastore (at least during inference time)



Retrieval-based language models

Retrieval-based NLP

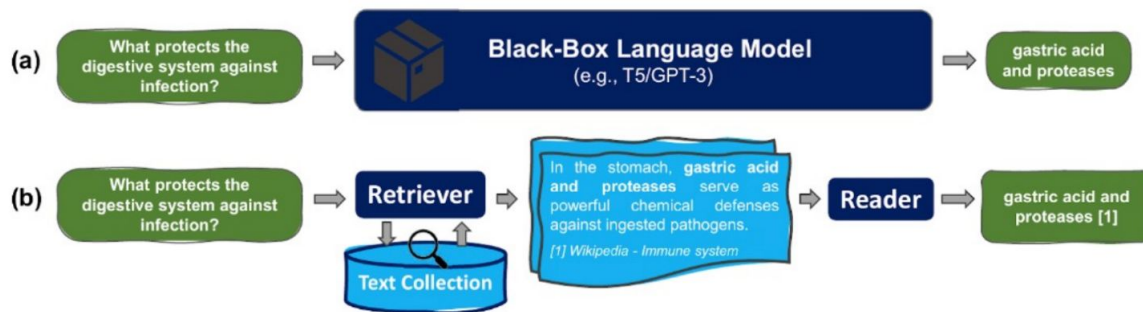
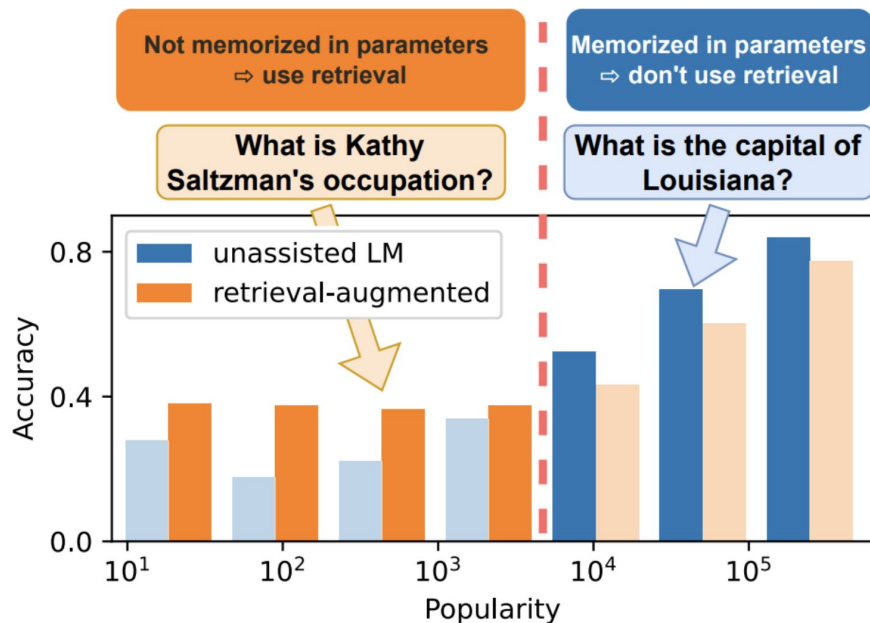


Figure 1: An illustration comparing (a) black-box language models and (b) retrieval-oriented NLP models, the paradigm this post advocates for.

Why retrieval-based language models?

- LLMs cannot memorize all (long-tail) knowledge in their parameters



Why retrieval-based language models?

- LLMs' knowledge is easily outdated and hard to update

Which team won the current champions league?

Paris Saint-Germain (PSG) are the current UEFA Champions League champions! They secured their **first-ever Champions League title** on **May 31, 2025**, with a stunning **5–0** victory over Inter Milan in the final at Munich's Allianz Arena [en.wikipedia.org · 15](#).

Key highlights:

- PSG dominated with a record-breaking winning margin for a Champions final [as.com](#).
- Under the guidance of manager **Luis Enrique**, they achieved a historic continental treble: Ligue 1, Coupe de France, and the Champions League [en.wikipedia.org · 15](#).
- Young talents like **Désiré Doué** starred—Doué scored twice, assisted once, and was named man of the match in Munich [en.wikipedia.org · 5](#).

Would you like insights on standout players, full match stats, or PSG's journey through the tournament?

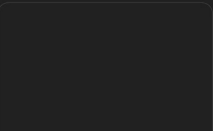
Top Champions League news



[reuters.com](#)

Mbappe unfazed by PSG winning Champions League without him

5일 전



[bavarianfootballworks.com](#)

Paris Saint-Germain vs. Inter Milan: CONGRATULATIONS TO PSG AND LUIS ENRIQUE, CHAMPIONS LEAGUE WINNERS 2024/25!

12일 전



[fourfourtwo.com](#)

Ousmane Dembele's Ballon d'Or and Simone Inzaghi leaving? What the winners and losers said after the Champions League final

12일 전

Why retrieval-based language models?

- Private issues

2025 사내 AI 챗봇 도입의 모든 것, 대기업 활용 사례부터 구축 방법까지

국내 주요 기업들의 AI 챗봇 활용 현황부터 자체 구축 방법까지, 성공적인 사내 AI 챗봇 도입을 위한 모든 것을 알아봅니다. 실제 활용 사례를 통해 기업에 최적화된 AI 챗봇 구축 방안을 제시합니다.



Retriever

- **Sparse retriever**
 - TF-IDF, BM-25
 - Basis - Lexical (word-level matching)
- **Dense retriever (embedding based retriever)**
 - DPR, contriever, openai-embeddings, etc
 - Basis - Semantic similarity (mean-based matching)

$$w_{x,y} = \text{tf}_{x,y} \times \log \left(\frac{N}{\text{df}_x} \right)$$

TF-IDF

Term x within document y

$\text{tf}_{x,y}$ = frequency of x in y

df_x = number of documents containing x

N = total number of documents

Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models

Yanzhao Zhang* Mingxin Li* Dingkun Long* Xin Zhang*
Huan Lin Baosong Yang Pengjun Xie An Yang
Dayiheng Liu Junyang Lin Fei Huang Jingren Zhou
Tongyi Lab Alibaba Group



<https://huggingface.co/Qwen>



<https://modelscope.cn/organization/qwen>



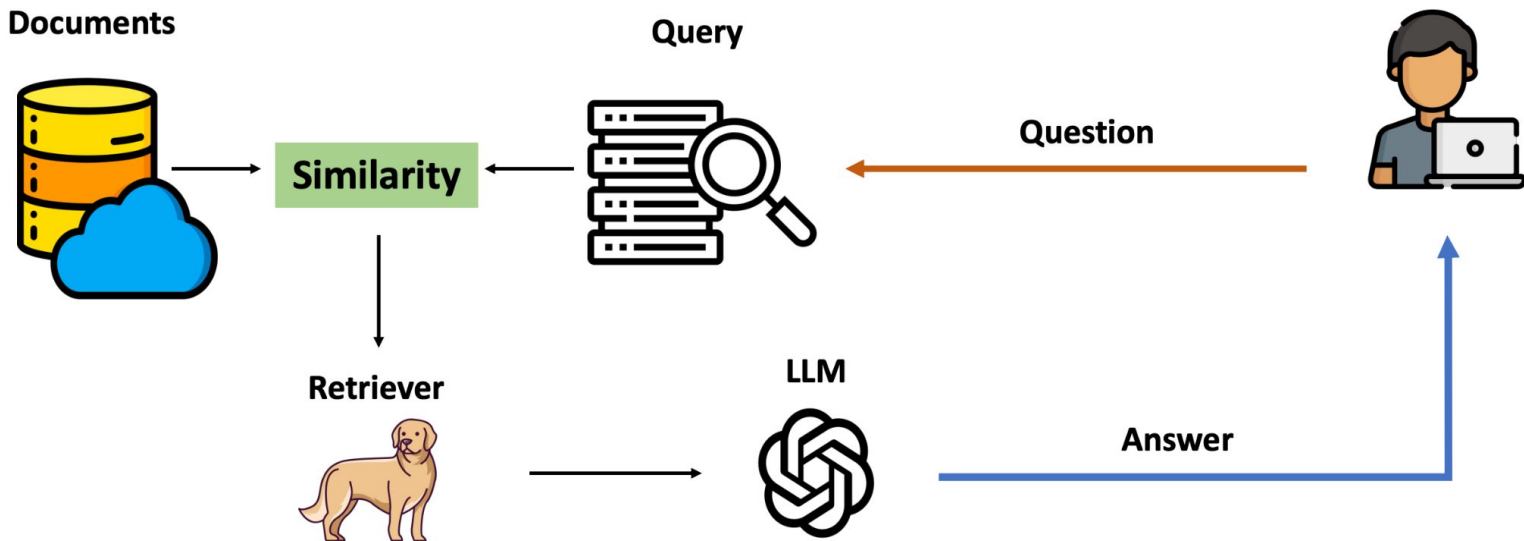
<https://github.com/QwenLM/Qwen3-Embedding>

Vector store

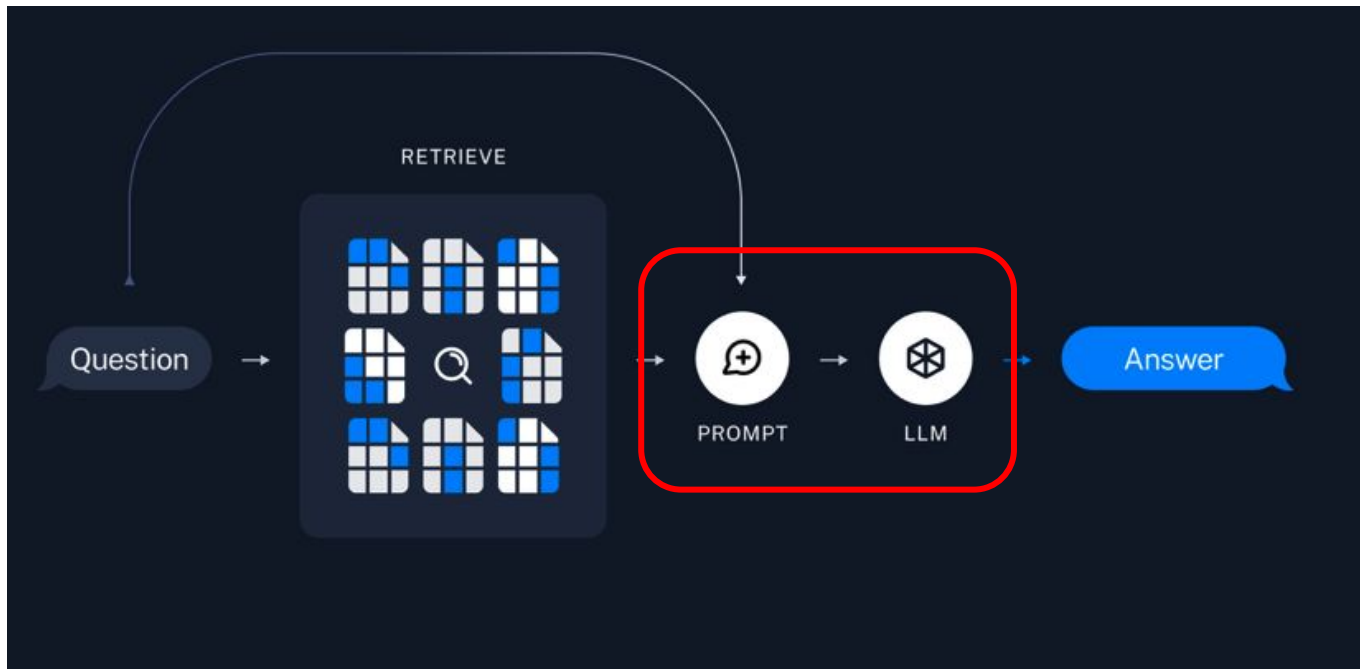
- Database optimized for storing and searching high-dimensional embeddings
- Chroma, FAISS



Retrieval Augmented Generation (RAG)



Retrieval Augmented Generation (RAG)



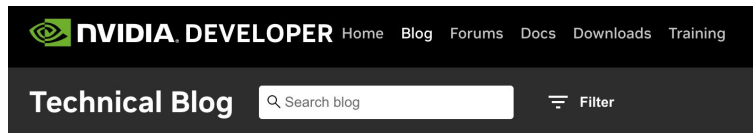
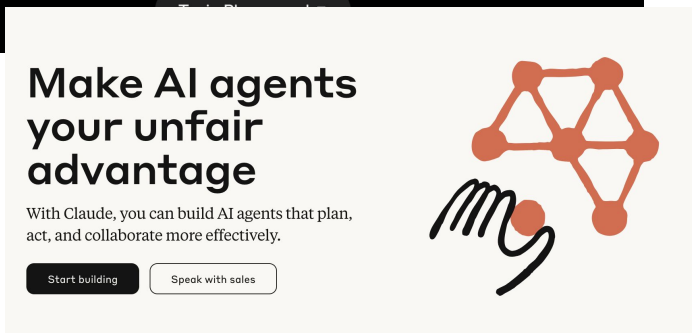
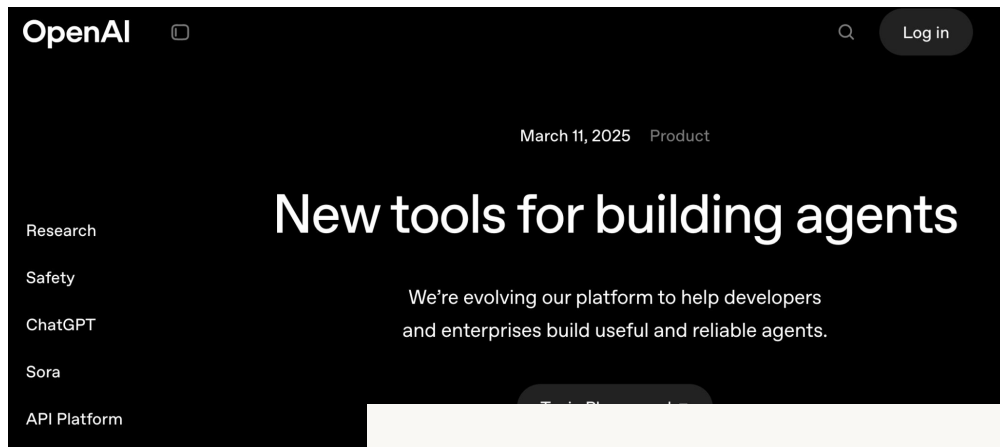
Build a RAG system

- Basic RAG application
- Build a Samsung Chatbot
 - 2024 삼성 지속가능경영보고서 기반의 chatbot
(https://www.samsung.com/sec/sustainability/media/pdf/Samsung_Electronics_Sustainability_Report_2024_KOR.pdf)
- Build a Debate workflow system
 - Agent 1 – 특정 주제에 대해 찬성
 - Agent 2 – 특정 주제에 대해 반대

LLM Agents

AI Agents

- Recently AI/LLM agents are getting a lot of attention



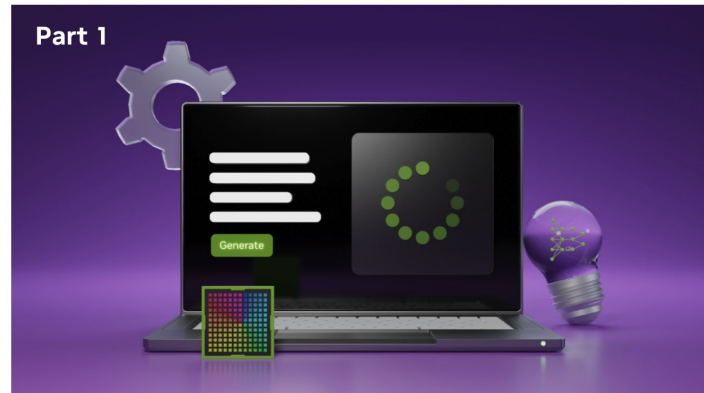
Generative AI

English ▾

Introduction to LLM Agents

Nov 30, 2023

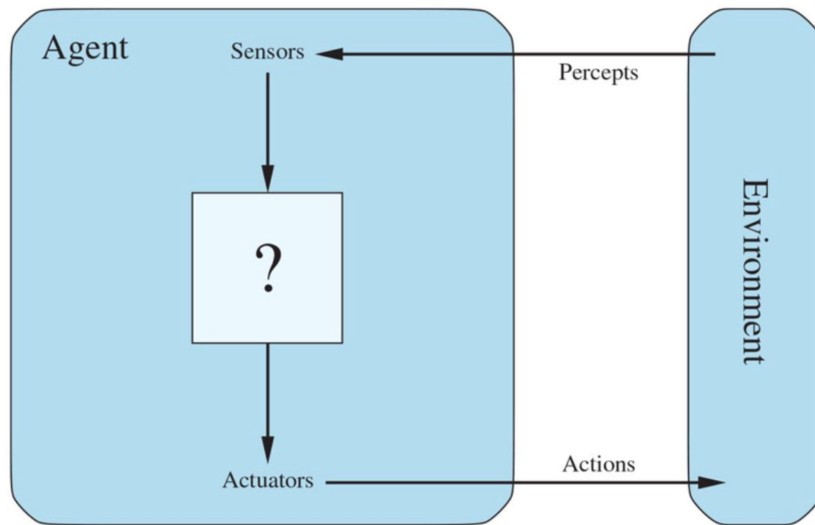
+85 Like Discuss (0)

By [Tanay Varshney](#)

Agent

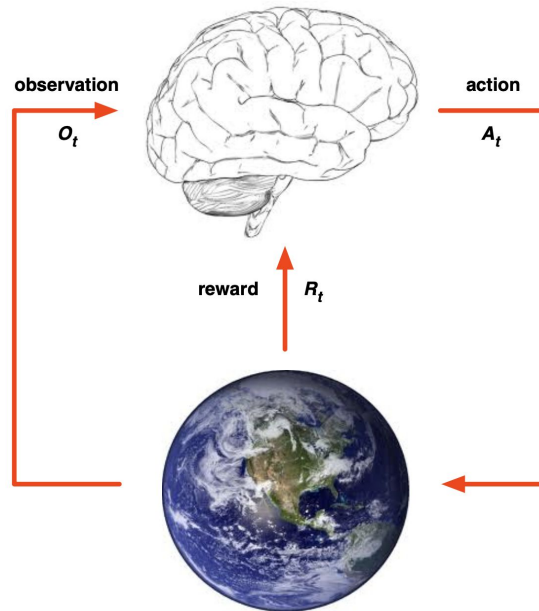
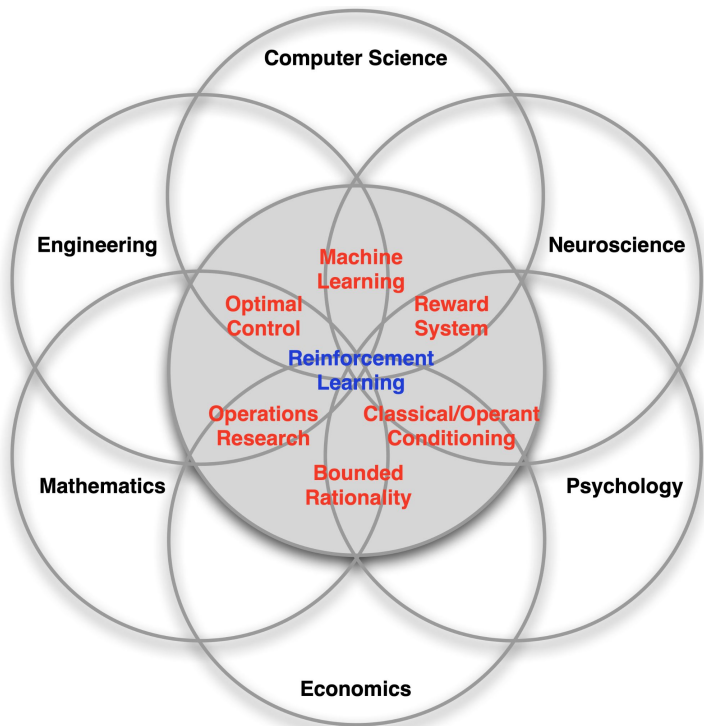
- **What is the Agent?**

- Anything that can be viewed as **perceiving** its environment through sensors and **acting upon** that **environment** through actuators – Russell & Norvig, *AI: A Modern Approach* (2020)



Agent

- Reinforcement Learning - Science of Decision Making



LLM Agent (or Language Agent)

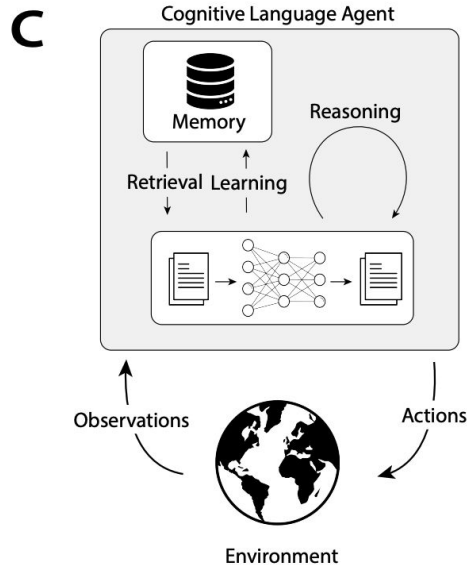
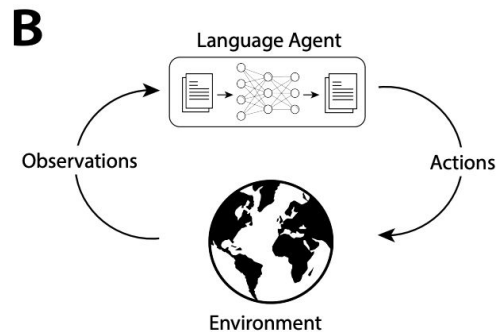
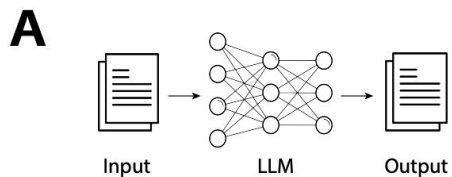
- What is the LLM Agent?

- **LLM-first view**: make LLM into an agent
 - e.g., RAG system -> RAG agent, Counselor agent
- **Agent-first view**: integrate LLMs into AI agents so they can use language for reasoning and communication
 - e.g., Embodied agent, GUI agent, Web agent, Code agent, etc

LLM Agent

- **Key concepts for LLM agents**

- Reasoning, Memory, Planning



Agent Design

- How can we design an agent?
 - <https://www.anthropic.com/engineering/building-effective-agents>



Workflow vs Agent

- **Agentic systems**

- **Workflows:** systems where LLMs and tools are orchestrated through predefined code paths
- **Agents:** systems where **LLMs dynamically direct their own processes** and tool usage, maintaining control over how they accomplish tasks

Workflow vs Agent

- **Agentic systems**

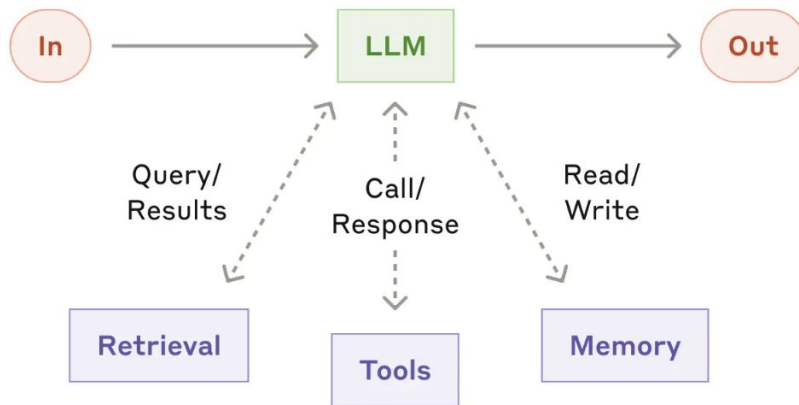
- **Workflows:** systems where LLMs and tools are orchestrated through predefined code paths
 - Predictability and consistency for well-defined tasks
- **Agents:** systems where **LLMs dynamically direct their own processes** and tool usage, maintaining control over how they accomplish tasks
 - Flexibility and model-driven **decision-making** are needed at scale

- **Note**

- For many applications, optimizing single LLM with retrieval and in-context examples is usually enough

Agent Design

- Building block: **The augmented LLM**
 - one way to implement this: **Model Context Protocol (MCP)**



Agent Design

Tools

- An agent can choose its actions (response) based on the tools it has access to
- It is important to define the available tools that the agent can utilize for the target task

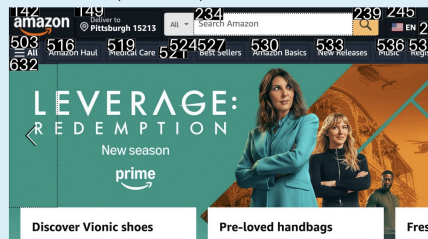
Table 4: List of available agent motor skills.

Skill	Description
Navigate	Used for navigating to an entity. You must provide the name of the entity you want to navigate to. Example: Navigate[counter_22].
Pick	Used for picking up an object. You must provide the name of the object. Example: Pick[cup_1].
Place	Used for placing an object on a target location. Example: Place[book_0, on, table_2, None, None].
Open	Used for opening an articulated entity. Example: Open[chest_of_drawers_1].
Close	Used for closing an articulated entity. Example: Close[chest_of_drawers_1].
Explore	Doing exploration towards a target object or receptacle, you need to provide the name of the place you want to explore.
Wait	Used to make agent stay idle for some time. Example (Wait[])

User Instruction (I)

Buy Sony WH-1000XM4 in Amazon.

Observation ($o^{\text{img}}, o^{\text{txt}}$)



[142] link 'Amazon', url='https://www.amazon.com/ref=nav_logo'
 [149] button 'Deliver to Pittsburgh 15213\@u200c'
 [234] searchbox 'Search Amazon', autocomplete='list', ...
 [239] button 'Go'
 [245] link 'Choose a language for shopping in Amazon United ..
 [251] button 'Expand to Change Language or Country'
 [503] button 'Open All Categories Menu', expanded=False
 ...

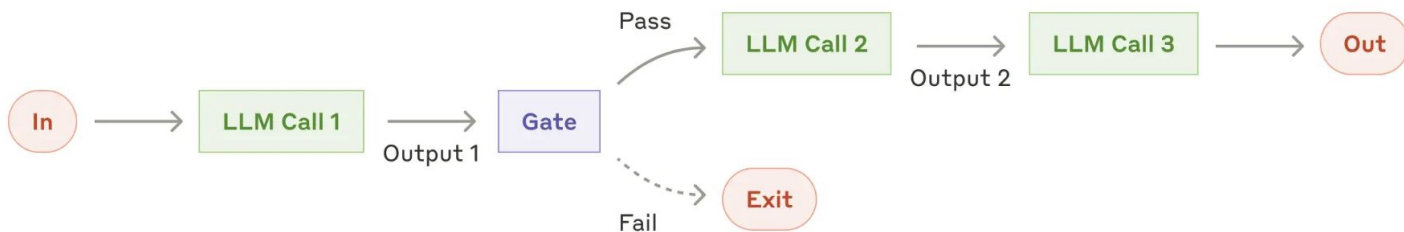
Action (a)

fill(234, "Sony WH-1000XM4")

Workflow

● Workflow 1: Prompt Chaining

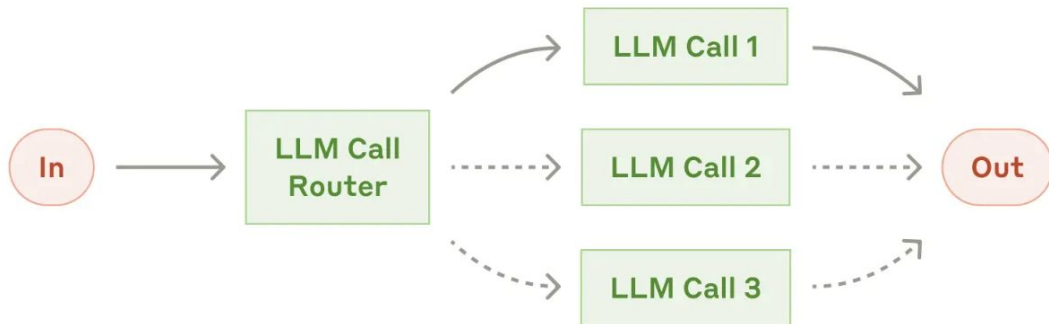
- Decomposes a task into sequence of steps
- when to use?
 - tasks that can be easily and clearly decomposed into fixed subtasks
 - e.g., writing an outline of a document, checklist



Workflow

• Workflow 2: Routing

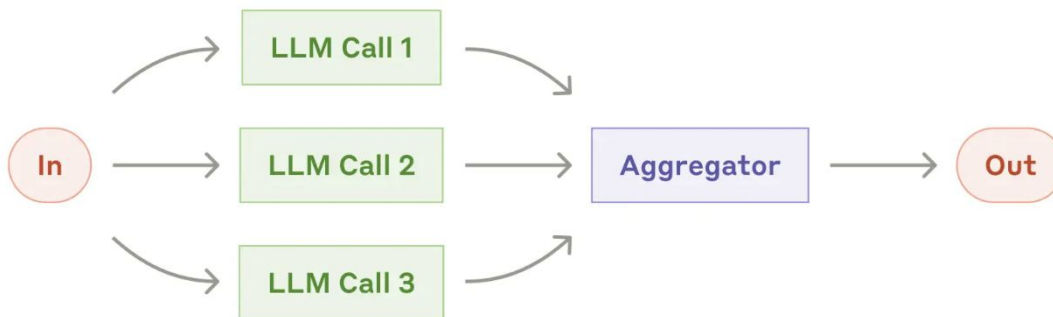
- Classifies an input and directs it to a specialized follow-up task
- This allows building more specialized prompts
- When to use?
 - Customer service queries



Workflow

● Workflow 3: Parallelization

- Work simultaneously on a task with output aggregation
 - Sectioning: breaking a task into independent subtasks run in parallel
 - Voting: Running the same task multiple times to get diverse outputs
- When to use?
 - Sectioning -> when divided subtasks can be parallelized for speed
 - Voting -> complex tasks with multiple consideration



Workflow

- **Workflow 4: Orchestrator-workers**

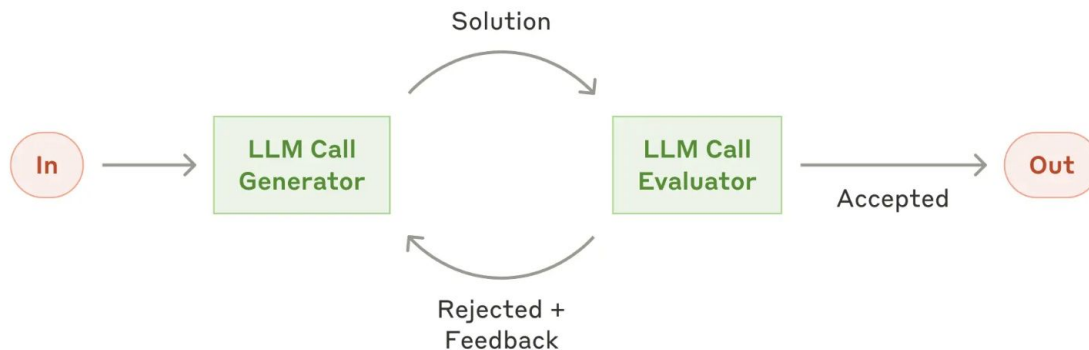
- Central LLM dynamically breaks down tasks, delegates them to worker LLMs, and synthesizes their results
- When to use?
 - complex tasks where you can't predict the subtasks needed
- Difference between 'parallelization'
 - **Flexibility**



Workflow

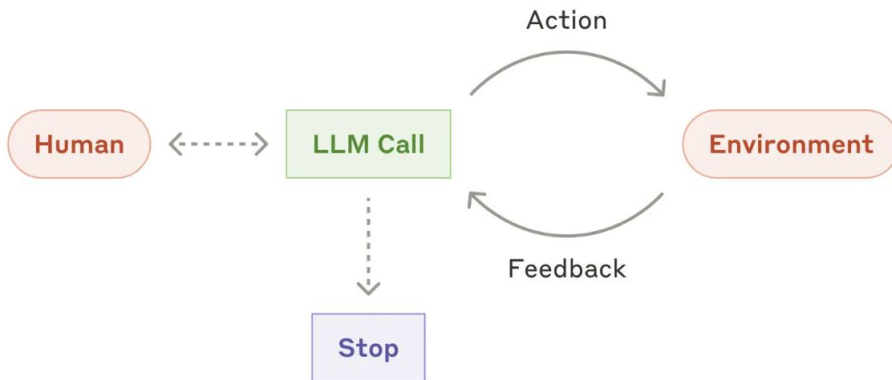
- **Workflow 5: Evaluator-optimizer**

- One LLM call generates a response while another provides evaluation and feedback in a loop
- When to use?
 - when we have clear **evaluation criteria**
 - when iterative refinement provides measurable values



Agent

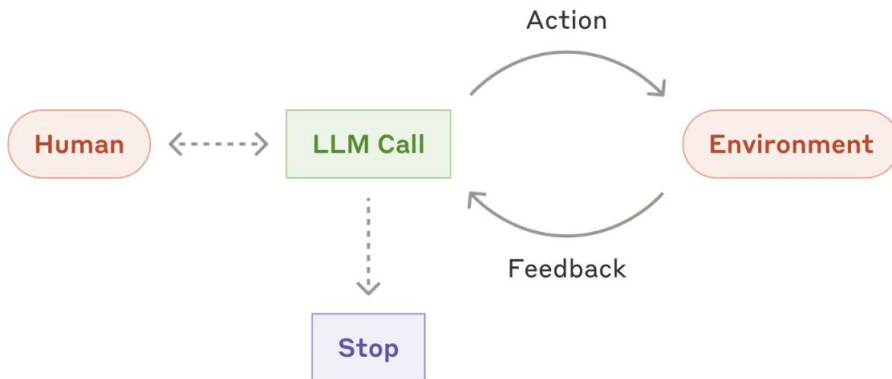
- **Autonomous agent**
 - Key capabilities
 - Understanding complex inputs
 - Engaging in reasoning and planning
 - Using tools reliably
 - Recover from errors



Agent

- **Autonomous agent**

- It's important for the agents to gain “**ground truth**” from the environment at each step
- When to use?
 - For open-ended problems
 - When you have trust in LLMs' decision-making ability

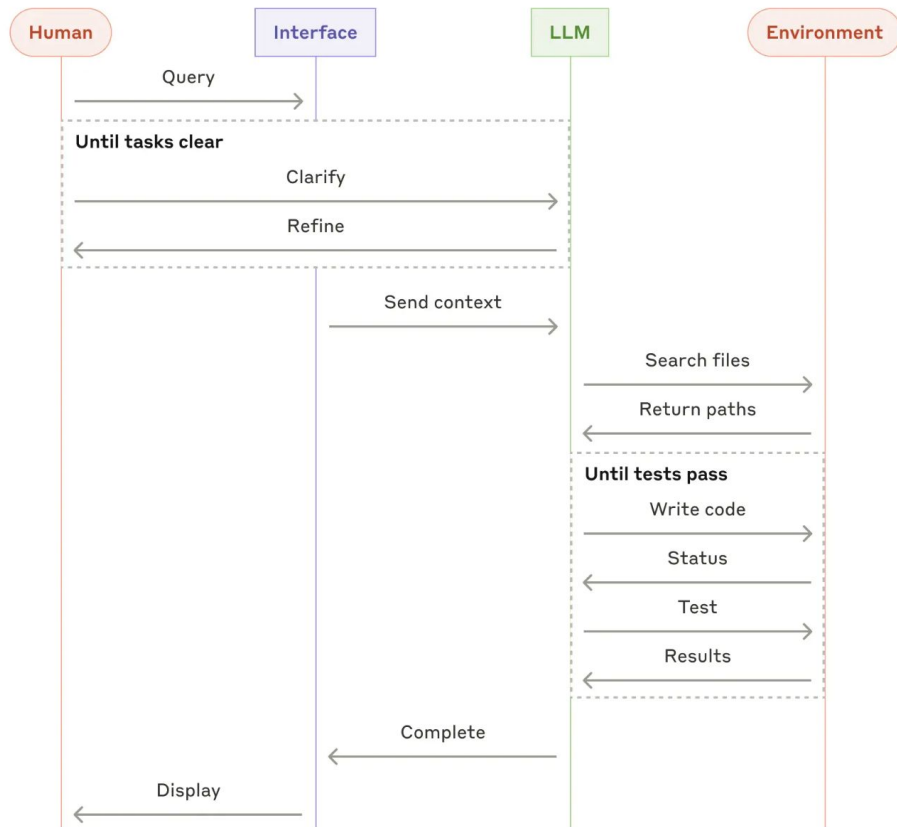


Agent

- **Coding agent high-level flow**

- SWE-bench

(<https://www.swebench.com/>)

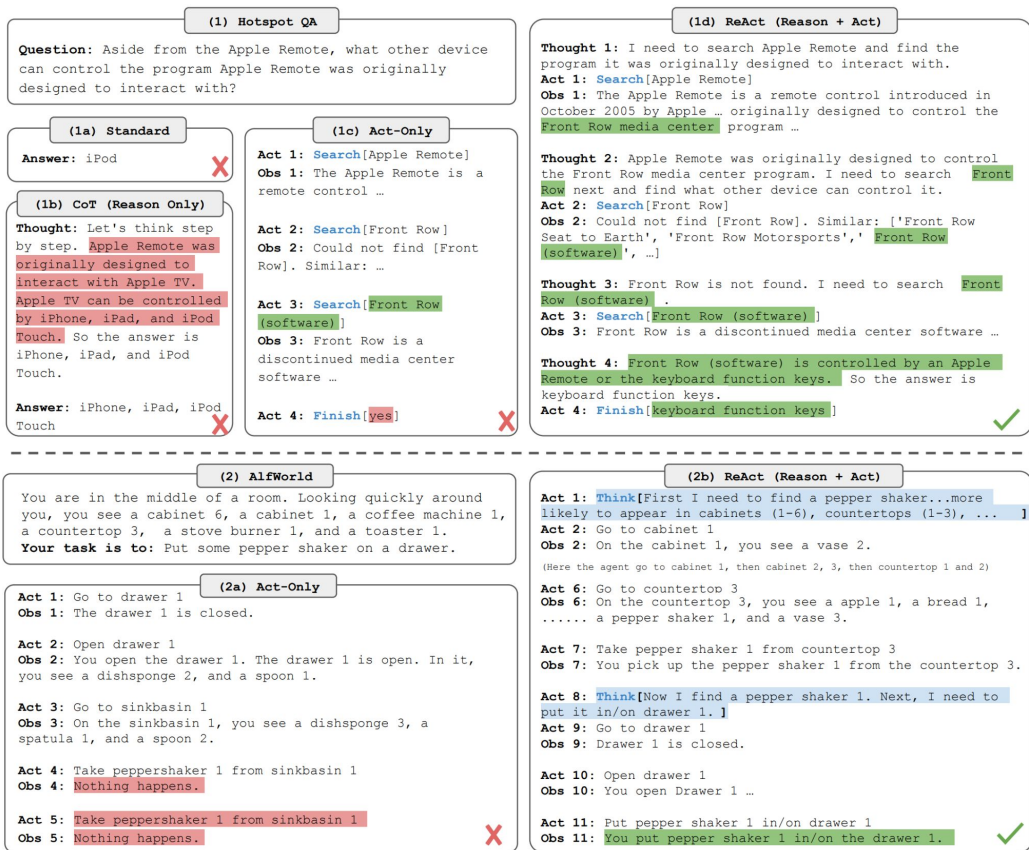


ReACT

• ReACT: Reason + Act

(<https://arxiv.org/abs/2210.03629>)

Base prompt format for current LLM agent



Alfworld

- Alfworld - TextWorld based embodied agent simulator (<https://alfworld.github.io/>)

TextWorld



```
Welcome!
You are in the middle of the room.
Looking around you, you see
a diningtable, a stove,
a microwave, and a cabinet.

Your task is to:
Put a pan on the diningtable.

> goto the cabinet
You arrive at the cabinet.
The cabinet is closed.

> open the cabinet
The cabinet is empty.

> goto the stove
You arrive at the stove. Near the
stove, you see a pan, a pot,
a bread loaf, a lettuce,
and a winebottle.

> take the pan from the stove
You take the pan from the stove.

> goto the diningtable
You arrive at the diningtable.

> put the pan on the diningtable
You put the pan on the
diningtable.
```

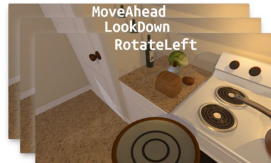
Embodied



RotateRight
LookDown
MoveAhead



Pickup



MoveAhead
LookDown
RotateLeft



Put

Hands On

- **w/ ReACT format prompting**
 - QA agent (with access to wikipedia)